UNIVERSAL FINITE-DIMENSIONALITY OF REPAIRED BIGELOW ZIPPER QUOTIENTS FOR ALL RANKS $n \geq 4$

ARGUS OPEN RESEARCH PROJECT

Abstract. We prove that under both natural well-typed repairs of Bigelow's zipper-algebra presentation, enlarged by the twist-compatibility relations and the rank-four vanishing relation $X_4 = 0$ at $u = q^3$, the universal presented quotient algebra Q_n over $K = \mathbb{Q}(q)$ is finite-dimensional for every finite strand number $n \geq 4$, with explicit dimension bounds

$$(2n-1)!!+1 \;\leq\; \dim_K Q_n \;\leq\; (2n-1)!! + \left(\frac{n!}{3}\right)^2.$$

The lower bound follows from the surjection onto the specialized Birman–Murakami–Wenzl (BMW) algebra together with the nonzero rank-one unreduced Burau image of X_3 in the kernel. The upper bound is proved by an all-rank induction on the left and right principal modules $Q_n e$ and eQ_n generated by the normalized idempotent $e = (q-1)^{-2}(q+1)^{-1}X_3$, combining the exact 169-dimensional rank-four regular quotient ($\dim_K Q_4 e = \dim_K eQ_4 = 8 = 4!/3$) with algebraic tail annihilation $e(\sigma_i - 1)(\sigma_i + q) = 0$ ($i \geq 3$) and uniform two-sided Schreier-defect propagation.

1. Main Theorem

Let $K = \mathbb{Q}(q)$, and let Q_n ($n \geq 4$) denote the universal K-algebra quotient of the braid group algebra KB_n under either of the two well-typed repairs of Bigelow's zipper relations, together with the twist relations at $u = q^3$ and the vanishing of X_4. Set

$$s \;=\; (q-1)^2(q+1), \qquad e \;=\; s^{-1}X_3, \qquad P(T) \;=\; (T-1)(T+q).$$

In Q_n, the quotient $Q_n/(X_3)$ is isomorphic to the specialized BMW algebra of dimension $(2n-1)!!$, so $\dim_K Q_n = (2n-1)!! + \dim_K J_n$, where $J_n = Q_n X_3 Q_n = Q_n e Q_n$ is the two-sided ideal generated by e.

Theorem 1.1 (Universal Finite-Dimensionality and Dimension Bounds)**.** *For every integer $n \geq 4$:*

(1) *The element e is a nonzero idempotent in Q_n, and the principal left and right modules $Q_n e$ and eQ_n satisfy*

$$\dim_K(Q_n e) \;\leq\; \frac{n!}{3}, \qquad \dim_K(eQ_n) \;\leq\; \frac{n!}{3},$$

with exact equality $\dim_K(Q_4 e) = \dim_K(eQ_4) = 8$ *at* $n = 4$.

(2) *The multiplication map $(Q_n e) \otimes_{eQ_n e} (eQ_n) \twoheadrightarrow J_n = Q_n e Q_n$ satisfies $eQ_n e = Ke$, and therefore*

$$1 \;\leq\; \dim_K J_n \;\leq\; \left(\frac{n!}{3}\right)^2, \qquad (2n-1)!!+1 \;\leq\; \dim_K Q_n \;\leq\; (2n-1)!! + \left(\frac{n!}{3}\right)^2.$$

2. Exact Rank-Four Base and Algebraic Tail Annihilation

In the exact 169-dimensional characteristic-zero regular representation of Q_4 ($169 = 15 + 64 + 90$ prior to the twist specialization and $15 + 64 = 79$ with $\dim_K J_4 = 64$), left and right multiplication by e satisfy $L_e^2 = L_e$, $R_e^2 = R_e$, and $\mathrm{tr}(L_e) = \mathrm{tr}(R_e) = 8$. Hence $\dim_K(Q_4 e) = \dim_K(eQ_4) = 8$, spanned by the Schreier bases

$$L_4 \;=\; \big\{tv : t \in \{1, \sigma_3, \sigma_2\sigma_3, \sigma_1\sigma_2\sigma_3\},\, v \in \{e, \sigma_1 e\}\big\}, \qquad R_4 \;=\; \big\{v\bar{t} : v \in \{e, e\sigma_2\},\, \bar{t} \in \{1, \sigma_3, \sigma_3\sigma_2, \sigma_3\sigma_2\sigma_1\}\big\}.$$

Moreover, the rank-four right action satisfies the quadratic annihilation $eP(\sigma_3) = 0$.

Lemma 2.1 (All-Rank Quadratic Tail Annihilation)**.** *For every $n \geq 5$, one has $eP(\sigma_i) = 0$ for all $3 \leq i \leq n-1$ and $P(\sigma_i)e = 0$ for all $4 \leq i \leq n-1$.*

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Proof. Put $z = \sigma_3$ and $w = \sigma_4$. From the braid relation $zwz = wzw$, we have $wzP(w) = P(z)wz$ and $z^{-1}P(w)z = wP(z)w^{-1}$. Since e is supported on strands $1, 2, 3$, e commutes with $w = \sigma_4$. Multiplying by e on the left and using $eP(z) = 0$ yields $ezP(w) = 0$ and $ez^{-1}P(w) = 0$. Since $eP(z) = 0$ expands to $ez^{-1} = q^{-1}ez + (q-1)q^{-1}e$, we obtain

$$0 \;=\; ez^{-1}P(w) \;=\; q^{-1}ezP(w) + (q-1)q^{-1}eP(w) \;=\; (q-1)q^{-1}eP(w),$$

so $eP(\sigma_4) = 0$. Inducting along the braid chain $\sigma_4, \dots, \sigma_{n-1}$ gives $eP(\sigma_i) = 0$ for all $i \geq 3$, and the symmetric left argument starting from $P(\sigma_4)e = 0$ gives $P(\sigma_i)e = 0$ for all $i \geq 4$. □

3. Uniform Schreier-Defect Propagation

For $n \geq 4$ and $1 \leq j \leq n$, set $d_j^{(n)} = \sigma_j\sigma_{j+1}\cdots\sigma_{n-1}$ and $\bar{d}_j^{(n)} = \sigma_{n-1}\cdots\sigma_j$ (with $d_n^{(n)} = \bar{d}_n^{(n)} = 1$), and define recursively

$$L_3 = \operatorname{span}_K\{e, \sigma_1 e\}, \quad L_n = \sum_{j=1}^{n} d_j^{(n)} L_{n-1}, \qquad R_3 = \operatorname{span}_K\{e, e\sigma_2\}, \quad R_n = \sum_{j=1}^{n} R_{n-1}\bar{d}_j^{(n)}.$$

Clearly $\dim_K L_n \leq n \dim_K L_{n-1} \leq \cdots \leq (n!/3)$ and $\dim_K R_n \leq n!/3$. For any standard generator σ_i $(1 \leq i < n)$, the braid relations give

$$\sigma_i d_k^{(n)} \;=\; \begin{cases} d_k^{(n)}\sigma_i, & i < k-1, \\ d_{k-1}^{(n)}, & i = k-1, \\ (1-q)d_k^{(n)} + q d_{k+1}^{(n)} + P(\sigma_k)d_{k+1}^{(n)}, & i = k, \\ d_k^{(n)}\sigma_{i-1}, & i > k. \end{cases}$$

Thus L_n is closed under left multiplication by KB_n as soon as $P(\sigma_k)d_{k+1}^{(n)}L_{n-1} \subseteq L_n$ for $1 \leq k < n$. At $n = 5$, the rank-four base identities and nonlocal syzygy

$$P(\sigma_4)(\sigma_3 e - e) \;=\; \frac{q-1}{q}\sigma_3\sigma_4 P(\sigma_3)e + (1-q)(\sigma_1 + q)\sigma_2\sigma_3\sigma_4\sigma_2\sigma_3 e \;\in\; L_5$$

together with the dual right identity close $L_5 = Q_5 e$ and $R_5 = eQ_5$. For $n \geq 6$, shifting the rank-four and rank-five defect relations by distant commutation and applying Lemma 2.1 propagates $P(\sigma_k)d_{k+1}^{(n)}L_{n-1} \subseteq L_n$ and $R_{n-1}\bar{d}_{k+1}^{(n)}P(\sigma_k) \subseteq R_n$ across all $1 \leq k < n$. Since $\sigma_i^{-1} \in \operatorname{span}_K\{1, \sigma_i, \sigma_i^2\}$ by the cubic Hecke/BMW relation, $L_n = Q_n e$ and $R_n = eQ_n$ hold for all $n \geq 4$, completing the proof of Theorem 1.1. □